SEC & Big Ten College Football Analytics Dashboard

Updated Data Source Plan
Developer: Matt Kurtz
Course: MSBA-692 Data Engineering
Technology Stack: Python, Requests, Pandas, SQLAlchemy, Supabase PostgreSQL, Dash, Plotly

1. Data Source Overview

The project uses the CollegeFootballData API as the primary source for college football team performance data. The selected dataset supports a focused MVP dashboard by providing season-level records that can be extracted, transformed, validated, stored, and visualized through an end-to-end data engineering pipeline.

The final project scope is intentionally limited to SEC and Big Ten teams. This keeps the dataset manageable while still allowing meaningful conference and team-level analysis.

2. Selected API Endpoint

Component	Final Project Selection
API Provider	CollegeFootballData API
Endpoint Used	Team Records endpoint
Season	2024
Dataset Scope	SEC and Big Ten teams
Returned Format	JSON response
Primary Fields Used	team name, conference, wins, losses

3. Business Relevance

The selected data source supports the dashboard's core business question: **How do SEC and Big Ten football teams compare by wins, losses, and winning percentage?** The endpoint provides clean season-level records that are appropriate for executive-style analytics, KPI reporting, and interactive dashboard filtering.

4. Data Acquisition Strategy

Data is extracted using Python's Requests library. The ETL script sends an authenticated request to the CollegeFootballData API, validates the API response, converts the returned JSON data into a Pandas DataFrame, and prepares the dataset for transformation and database loading.

- Authenticate using a CollegeFootballData API key.
- Request 2024 team records from the records endpoint.
- Validate the HTTP response before transformation.
- Parse the JSON response into a structured Pandas DataFrame.
- Filter the dataset to SEC and Big Ten teams only.

5. Transformation and Preparation Plan

After extraction, the pipeline applies transformation logic to make the data analytics-ready. The workflow standardizes fields, converts numeric values, calculates the derived win percentage metric, and removes records that fail validation checks.

- Standardize team and conference values.
- Convert wins and losses into numeric fields.
- Calculate win_percentage = wins / (wins + losses).
- Remove duplicate team records.
- Validate that wins and losses are non-negative.
- Export an analytics-ready CSV for backup and downstream use.

6. Storage and Analytics Use

Validated records are loaded into Supabase PostgreSQL using SQLAlchemy. The Dash application queries this database directly to populate KPI cards, charts, and the conference filter. This design demonstrates live database connectivity while keeping the MVP dashboard fast and reliable.

7. Risk Considerations

- API availability: handled through response validation and error logging.
- Schema changes: mitigated by validating required fields before loading.
- Duplicate records: prevented through primary key constraints and incremental loading logic.
- Scope creep: limited by focusing only on SEC and Big Ten team records.

8. Final Data Source Deliverable

The final dataset supports an interactive dashboard with Total Teams, Average Win Percentage, Wins by Team, Wins vs. Losses, and conference-level filtering. The data source is appropriate for the course project because it is public, structured, sports-related, manageable in scope, and rich enough for meaningful visual analytics.